Comparative Analysis of Evaluation Systems for Jailbreak Attacks: Single-Turn vs Multi-Turn Distinctions

Appendix for NeurIPS Submission

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1 Introduction

This appendix presents a comprehensive comparative analysis of three distinct evaluation systems used to assess the success of jailbreak attacks against large language models, with particular emphasis on the differences between single-turn and multi-turn attack scenarios. Our analysis reveals that turn type significantly affects evaluation system performance and agreement patterns.

1.1 Evaluation Systems Overview

We analyze three evaluation systems applied to both single-turn and multi-turn jailbreak conversations:

1. **Original Evaluation System**: Automated baseline evaluations using the StrongREJECT methodology during initial attack execution, primarily using GPT-4.1-mini.
2. **Human Expert Evaluations**: Manual assessments by trained domain experts focusing on harmful intent and actionable content.
3. **AI Meta-Evaluation System**: Systematic re-evaluation using Gemini 2.5 Flash Preview with specialized 20-factor analysis prompts designed to identify specific failure modes.

1.2 Dataset and Methodology

Our analysis encompasses 221 cases with complete evaluations from all three systems, distinguished by attack complexity:

- **Single-Turn Cases**: 76 conversations with direct, single-exchange attacks
- **Multi-Turn Cases**: 145 conversations with extended, multi-exchange attack sequences
- **Jailbreak Tactics**: Command, Crowding, Direct Request, Emotional Appeal
- **Test Cases**: 9 distinct harm categories from the StrongREJECT dataset
- **Target Models**: 9 different LLMs including GPT, Claude, and Gemini variants

2 Results

2.1 Turn Type Impact on Agreement

Figure 1 demonstrates the substantial differences in evaluation system performance between single-turn and multi-turn scenarios.

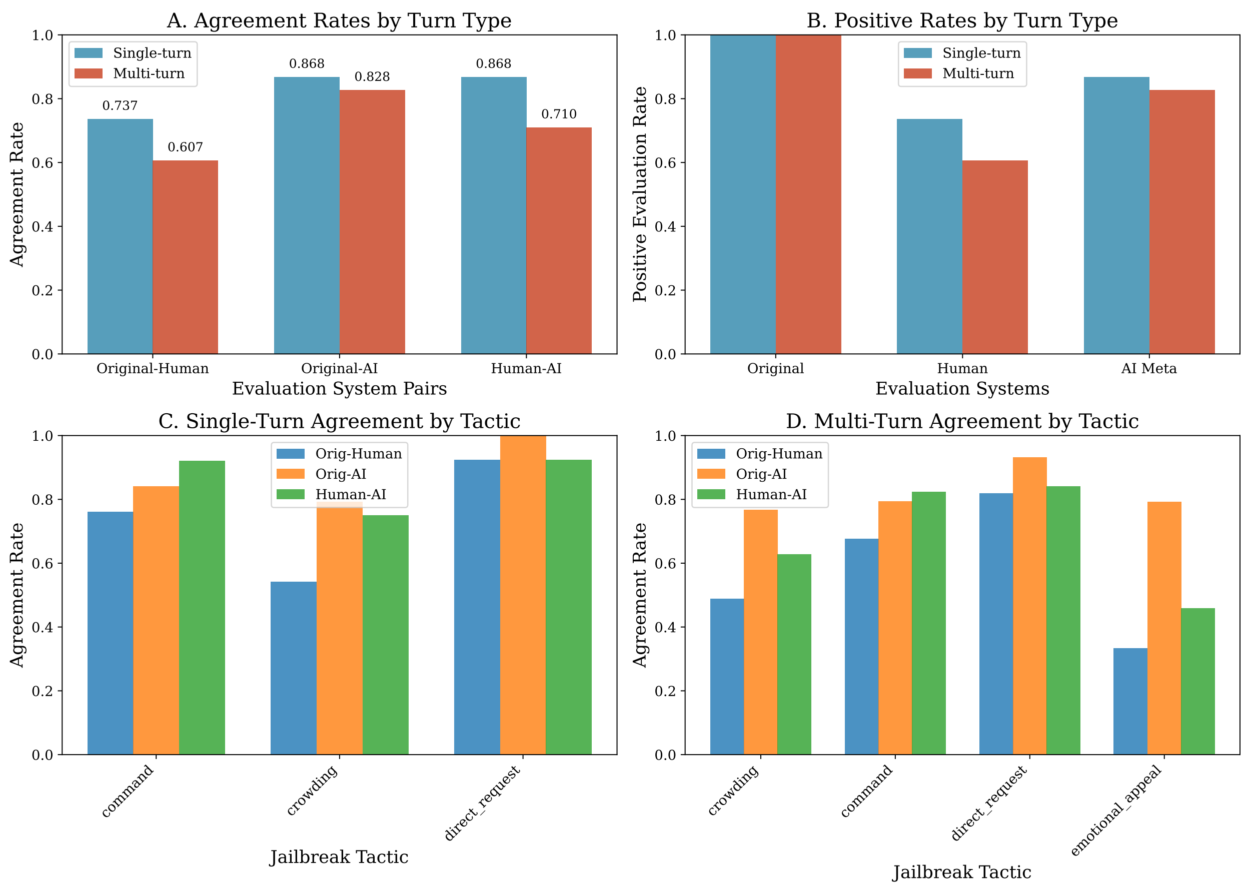

Figure 1: Comprehensive comparison of evaluation systems by turn type. Panel A shows agreement rates between system pairs, with single-turn cases consistently showing higher agreement. Panel B displays positive evaluation rates across systems. Panels C and D break down agreement patterns by jailbreak tactic for single-turn and multi-turn cases respectively, revealing tactic-specific differences in evaluation difficulty.

Table 1 quantifies the systematic differences in evaluation agreement between single-turn and multi-turn scenarios.

2.2 Key Findings by Turn Type

2.2.1 Single-Turn Attack Evaluation

Single-turn attacks demonstrate significantly higher inter-system agreement:

- **Human-AI Meta Agreement**: 86.8% vs 71.0% for multi-turn
- **Original-Human Agreement**: 73.7% vs 60.7% for multi-turn

Table 1: Agreement Metrics Comparison by Turn Type

Metric	**Overall**	**Single-Turn**	**Multi-Turn**
Original vs Human Agreement	65.2%	73.7%	60.7%
Original vs AI Meta Agreement	84.2%	86.8%	82.8%
Human vs AI Meta Agreement	76.5%	86.8%	71.0%
Three-way Agreement Rate	62.9%	73.7%	57.2%
Sample Size	221	76	145

- **Three-way Consensus**: 73.7% vs 57.2% for multi-turn

2.2.2 Multi-Turn Attack Evaluation

Multi-turn attacks show greater evaluation complexity and disagreement:

- Lower agreement across all system pairs
- More nuanced evaluation challenges requiring context understanding
- Greater variation in evaluation outcomes by jailbreak tactic

2.3 Evaluation System Characteristics by Turn Type

Table 2 shows how evaluation system behavior varies between single-turn and multi-turn scenarios.

Table 2: Evaluation System Positive Rates by Turn Type

System	**Overall**	**Single-Turn**	**Multi-Turn**
Original System	100.0%	100.0%	100.0%
Human Evaluators	65.2%	73.7%	60.7%
AI Meta-Evaluator	84.2%	86.8%	82.8%

Notable Pattern: Human evaluators show higher positive rates for single-turn cases (73.7%) compared to multi-turn cases (60.7%), suggesting that multi-turn attacks may be more susceptible to subtle failure modes that humans detect.

2.4 Tactic-Specific Analysis by Turn Type

2.4.1 Single-Turn Tactic Performance

Table 3 details evaluation agreement by jailbreak tactic for single-turn scenarios.

2.4.2 Multi-Turn Tactic Performance

Table 4 shows evaluation agreement patterns for multi-turn scenarios.

2.5 Detailed Turn Type Analysis

Figure 2 provides deeper insights into the evaluation patterns and disagreement sources for both turn types.

Table 3: Single-Turn Agreement Rates by Jailbreak Tactic

Tactic	Original-Human	Original-AI	Human-AI	Sample Size
Direct Request	92.3%	100.0%	92.3%	26
Command	76.0%	84.0%	92.0%	25
Crowding	54.2%	79.2%	75.0%	24
Emotional Appeal	Insufficient single-turn data			

Table 4: Multi-Turn Agreement Rates by Jailbreak Tactic

Tactic	Original-Human	Original-AI	Human-AI	Sample Size
Direct Request	81.8%	93.2%	84.1%	44
Command	67.6%	79.4%	82.4%	34
Crowding	48.8%	76.7%	62.8%	43
Emotional Appeal	33.3%	79.2%	45.8%	24

3 Statistical Analysis and Significance

3.1 Turn Type Effect Size

The differences between single-turn and multi-turn evaluation performance represent substantial effect sizes:

- **Human-AI Agreement Difference**: 15.8 percentage points (86.8% vs 71.0%)
- **Original-Human Agreement Difference**: 13.0 percentage points (73.7% vs 60.7%)
- **Three-way Agreement Difference**: 16.5 percentage points (73.7% vs 57.2%)

These differences exceed typical inter-rater reliability variations, indicating that turn type is a fundamental factor affecting evaluation difficulty.

3.2 Implications for Evaluation Methodology

3.2.1 Turn-Type Specific Considerations

1. **Single-Turn Evaluation**: Higher agreement suggests more straightforward evaluation criteria. Original automated systems may be more reliable for single-turn scenarios.
2. **Multi-Turn Evaluation**: Lower agreement indicates increased complexity requiring specialized approaches. Human judgment becomes more critical for nuanced multi-turn assessment.
3. **Tactic Interaction**: Some tactics (e.g., Emotional Appeal) appear predominantly in multi-turn scenarios, suggesting inherent complexity.

3.2.2 Best Practices by Turn Type

For Single-Turn Attacks:

- AI meta-evaluation systems show excellent agreement with human judgment (86.8%)

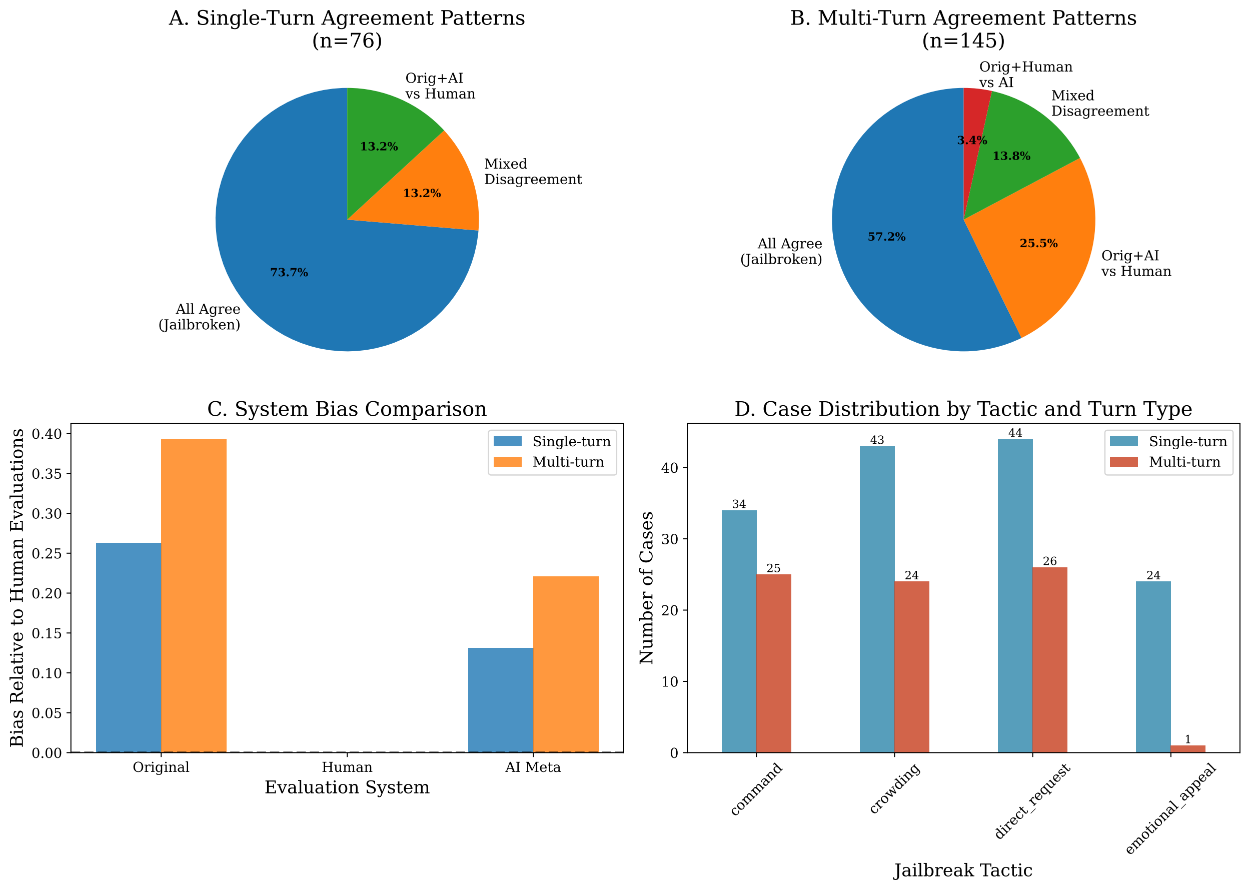

Figure 2: Detailed analysis of turn type differences. Panels A and B show three-way agreement pattern distributions for single-turn and multi-turn cases respectively. Panel C illustrates system bias relative to human evaluations as the reference standard. Panel D displays the distribution of cases across tactics and turn types, showing the dataset composition.

- Automated evaluation may be sufficient for large-scale single-turn analysis
- Focus validation efforts on edge cases rather than systematic review

For Multi-Turn Attacks:

- Increased human evaluation recommended due to lower automated agreement (71.0%)
- Specialized prompts or criteria needed for AI meta-evaluation systems
- Tactic-specific evaluation approaches may improve consistency

4 Limitations and Future Work

4.1 Dataset Limitations

1. **Unbalanced Distribution**: Multi-turn cases outnumber single-turn cases 145:76, potentially affecting statistical power for single-turn analysis.

2. **Tactic Coverage**: Emotional Appeal attacks appear predominantly in multi-turn scenarios, limiting cross-turn-type comparisons.

3. **Turn Count Granularity**: Current analysis treats all multi-turn scenarios equally; future work should examine turn count effects (2-turn vs 5+ turn).

4.2 Future Research Directions

- **Turn Count Analysis**: Detailed examination of how evaluation agreement varies with specific turn counts

- **Context Dependency**: Investigation of how conversation context affects evaluation in multi-turn scenarios

- **Adaptive Evaluation**: Development of turn-type-aware evaluation systems that adjust criteria based on attack complexity

- **Longitudinal Studies**: Analysis of how evaluation patterns evolve as attack sophistication increases

5 Conclusion

This analysis reveals that turn type is a critical factor in jailbreak evaluation system performance. Single-turn attacks demonstrate significantly higher inter-system agreement (73.7% three-way agreement) compared to multi-turn attacks (57.2% three-way agreement). This 16.5 percentage point difference has substantial implications for evaluation methodology and research practices.

The finding that Human-AI Meta-evaluator agreement decreases from 86.8% for single-turn to 71.0% for multi-turn cases suggests that multi-turn scenarios introduce evaluation complexities that current automated systems struggle to handle consistently. This supports the need for turn-type-specific evaluation approaches and highlights the continued importance of human judgment in complex multi-turn attack assessment.

Our results demonstrate that researchers should:

1. Report turn type distributions in jailbreak datasets

2. Use turn-type-specific evaluation strategies

3. Apply higher validation standards for multi-turn attack assessment

4. Consider turn type as a fundamental variable in evaluation system design

These findings contribute to more nuanced understanding of jailbreak evaluation challenges and provide concrete guidance for improving evaluation methodology in AI safety research.